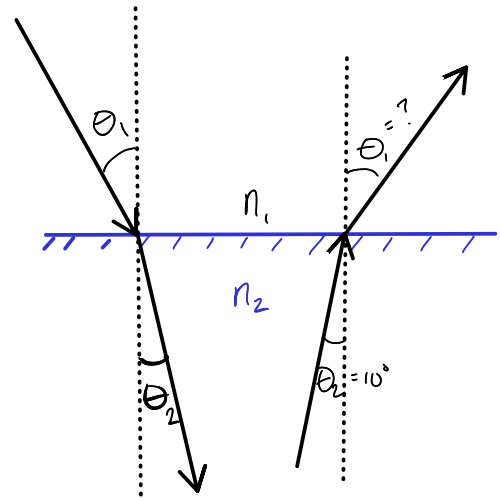
Week 13 covers sections 3,4, and 9 of chapter 23 in the textbook.
Topics include:

- refraction and Snell's Law
- dispersion (color separation by a prism)
- total internal reflection and critical angle
- image formation and converging and diverging lenses

1. Snell's law relates the angle of incidents and transmission to the indices of refraction of the material. Here is a formula for it:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

Let material 1 be air $n_1 = 1$ and material 2 be water $n_2 = 1.33$. When light is traveling from air to water at an incident angle of 30°, then what is the transmitted angle? Draw a picture of this. What angle of incidence would make the transmitted angle be 20°? If light were traveling from the water out to the air at an incident angle of 10° what angle would the light take in the air? What if this angle under the water were 20°?



$n_1 \sin\theta_1 = n_2 \sin\theta_2$

$\theta_2 = \sin^{-1}\left(\frac{n_1}{n_2}\sin\theta_1\right)$

$\theta_2 = \sin^{-1}\left(\frac{1}{1.33}\cdot\sin 30°\right)$

$\theta_2 = 22.1°$

$\theta_1 = \sin^{-1}\left(\frac{n_2}{n_1}\cdot\sin\theta_2\right)$

$= \sin^{-1}\left(\frac{1.33}{1}\sin(20°)\right)$

$\theta_1 = 27.1°$

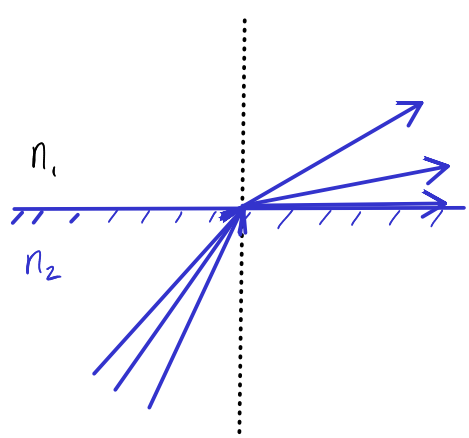
$\theta_1 = \sin^{-1}\left(\frac{1.33}{1}\cdot\sin(10°)\right)$

$\theta_1 = 13.35°$

$\theta_1 = \sin^{-1}\left(\frac{1.33}{1}\cdot\sin(20°)\right)$

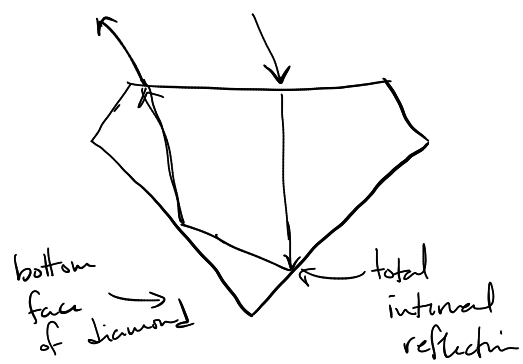
$\theta_1 = 27.1°$

2. Make a table of the incident angles and transmitted angles for light traveling from air into water. What angle does the transmitted angle approach? Then revert the direction of light and calculate a table for the light going from water into air. What happens?



air -->	water		water -->	air		water -->	air		water -->	air
n1	n2		n2	n1		n2	n1		n2	n1
0	0		0	0		41	60.75733		48.1	81.86377
10	7.502101		10	13.35313		42	62.86617		48.2	82.51679
20	14.90149		20	27.05765		43	65.10188		48.3	83.23069
30	22.08241		30	41.68233		44	67.50241		48.4	84.02723
40	28.90108		40	58.74952		45	70.12761		48.5	84.94535
50	35.16782		50	#NUM!		46	73.08195		48.6	86.06933
60	40.62813		60	#NUM!		47	76.58136		48.7	87.68138
70	44.95375		70	#NUM!		48	81.25795		48.8	#NUM!
80	47.77036		80	#NUM!		49	#NUM!		48.9	#NUM!
90	48.75347		90	#NUM!		50	#NUM!		49	#NUM!

3. What you have found is an angle for which the light does not emerge from the material, but rather reflects inside the material. This is known as *total internal reflection* and it is how fiber optic cables work. The angle for which this occurs is known as the *critical angle*. Now use Snell's Law to derive an expression for the critical angle in terms of the index of refraction of the two materials. Be careful about which index of refraction is for the material that the light is in, and which index of refraction that the light *does not* go into.

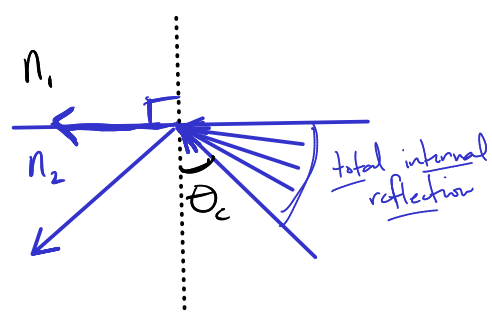


$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$\theta_1 = 90°$, $\sin 90° = 1$; $\theta_2 = \theta_c$

$$n_1 = n_2 \sin\theta_c$$

$$\theta_c = \sin^{-1}\left(\frac{n_1}{n_2}\right)$$



water → air

$$\theta_c = \sin^{-1}\left(\frac{1}{1.33}\right)$$

$$\theta_c = 48.753\ldots$$

4. What is the critical angle for light in glass surrounded by air? If you surround the glass with water, what does the critical angle become?

$n_{glass} = 1.5$ | $n_{water} = 1.33$ | $n_{air} = 1$ | $n_{diamond} = 2.41$

glass to air: $41.8°$

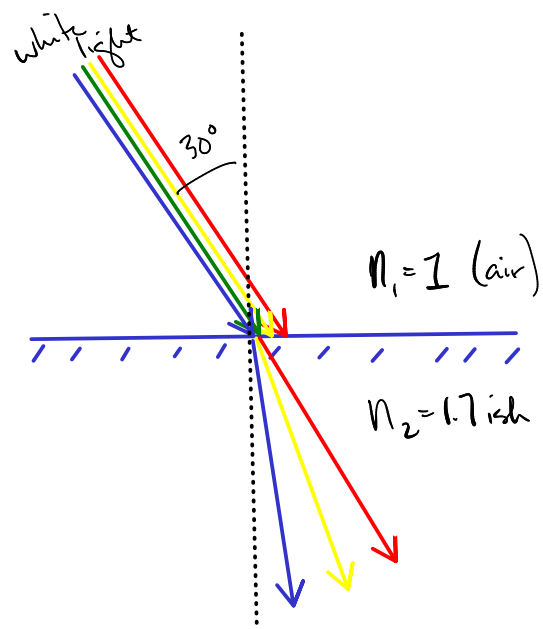
glass to water: $\theta_c = \sin^{-1}\left(\frac{1.33}{1.5}\right) = 62.5°$

diamond to air: $24.5°$

diamond to water: $33.5°$

5. I mentioned in the notes that white light is composed of all the colors in the visible spectrum at once and that the index of refraction of most materials (*slightly*) depends on the frequency of the light itself. Below is a table of three colors and the index of refraction in flint glass. What is the speed of these three wavelengths in flint glass? If light consisting of these colors arrives in a single beam at an incident angle of 30°, then what angle of transmission do each of these colors have?

λ (nm)	color	n (flint glass)	θ_t
486.1	blue	1.7328	16.77
589.2	yellow	1.7205	16.89
656.3	red	1.7076	17.03



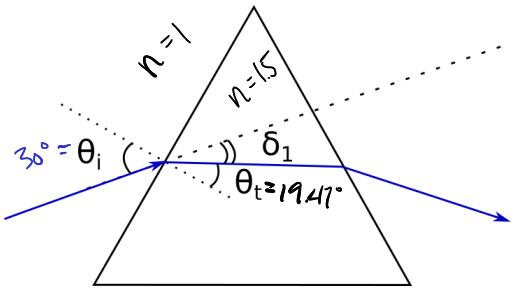
6. When light passes through a prism its path changes once on the way into the prism and then again on the way out. Assume the prism is an equilateral triangle of glass ($n = 1.5$).

(a) For an exterior incident angle of 30°, what is the transmitted angle in the prism?

$\rightarrow n_i \sin\theta_i = n_t \cdot \sin\theta_t$

$\theta_t = \sin^{-1}\left(\frac{n_i}{n_t}\sin\theta_i\right) = \sin^{-1}\left(\frac{1}{1.5}\sin(30°)\right) = 19.47°$

(b) Inside the prism, what is the *deviation* δ_1 of the light's path?



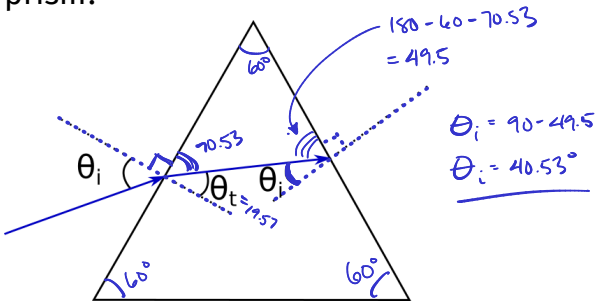
$$\theta_i = \delta_1 + \theta_t$$

$$\delta_1 = \theta_i - \theta_t$$

$$\delta_1 = 30° - 19.47°$$

$$\delta_1 = 10.53°$$

(c) What is the interior incident angle as the light is leaving the prism?



$$\theta_i = 90 - 49.5$$

$$\theta_i = 40.53°$$

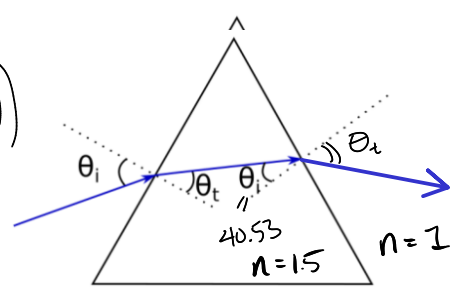
(d) What is the final transmitted angle as the light leaves the prism?

$$\theta_t = \sin^{-1}\left(\frac{n_i}{n_t} \cdot \sin\theta_i\right) = \sin^{-1}\left(\frac{1.5}{1} \cdot \sin(40.53)\right)$$

1.5

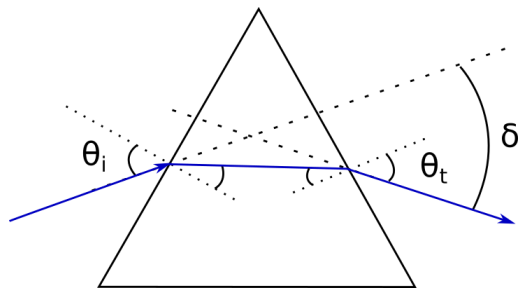
1

$$= 77.1°$$



(e) What is the second deviation of the light's path?

(f) What is the total deviation of the lights path?



(g) Using excel or google sheets, do this procedure with many exterior incident angles (all integer angles). Don't forget that excel's trig functions only accept radians and inverse trig functions will return angles in radians so you will have to include some conversion factors in the formula (π rad $=$ 180°). Plot the total deviation as a function incident angle and see if there are any interesting features.

7. For a converging lens with a focal length of +5 cm, plot the image distance as a function of object distance. Do a ray tracing for objects at three different object distances, one farther that 2 focal lengths away, one between 1 and 2 focal lengths away, and one less than a focal length away from the lens.

thin lens equation

$$\boxed{\frac{1}{p} + \frac{1}{q} = \frac{1}{f}}$$

magnification equation

$$\boxed{\frac{-q}{p} = m = \frac{h'}{h}}$$

$f = +5\text{ cm}$

$p = +15\text{ cm}$

$q = +7.5\text{ cm}$

$h = 2\text{ cm}$

$h' = -1\text{ cm}$

$m = -0.5$

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

$$\frac{1}{p} + \frac{1}{p} = \frac{1}{f}$$

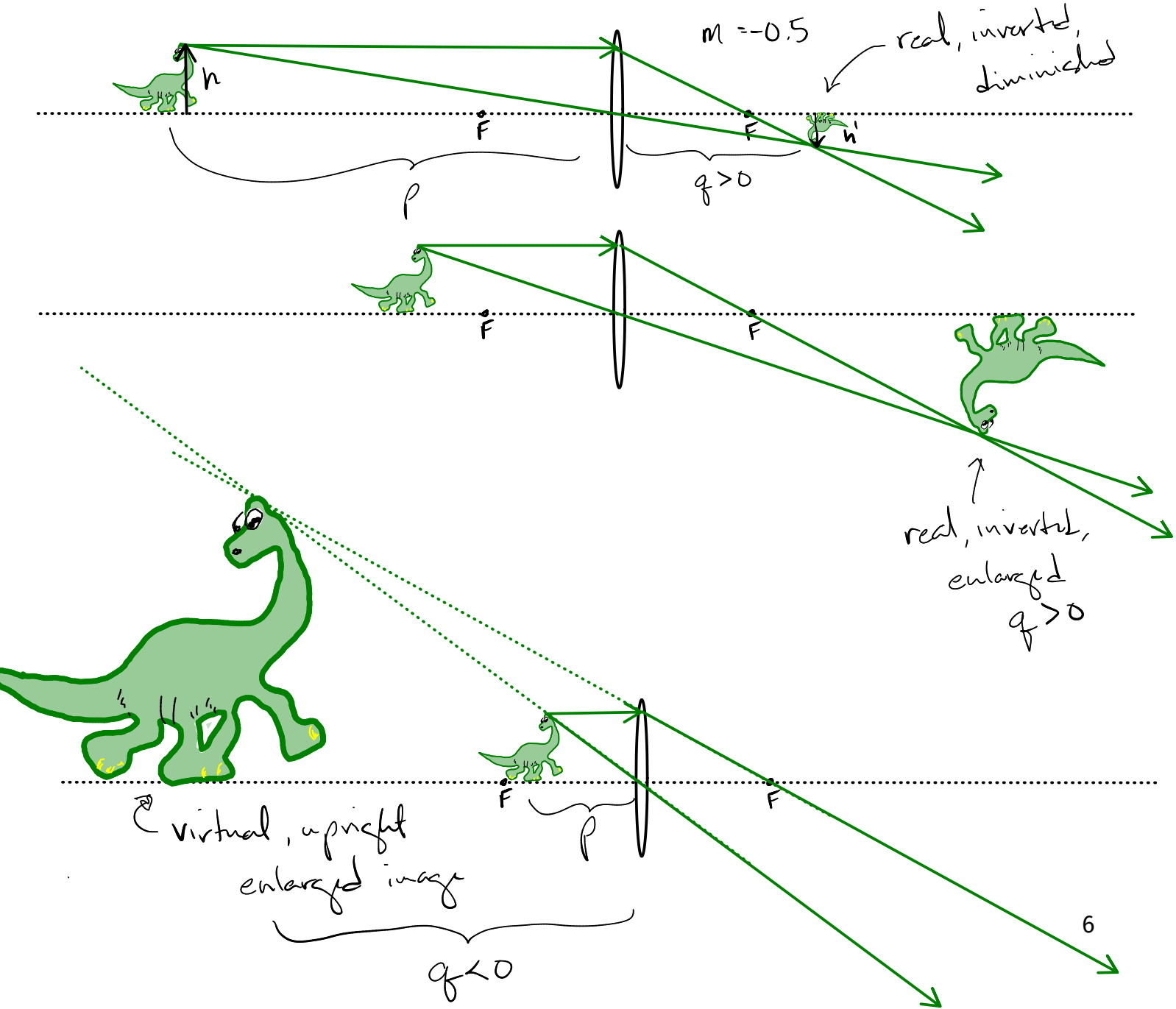
$$\frac{2}{p} = \frac{1}{f}$$

$$p = 2f$$

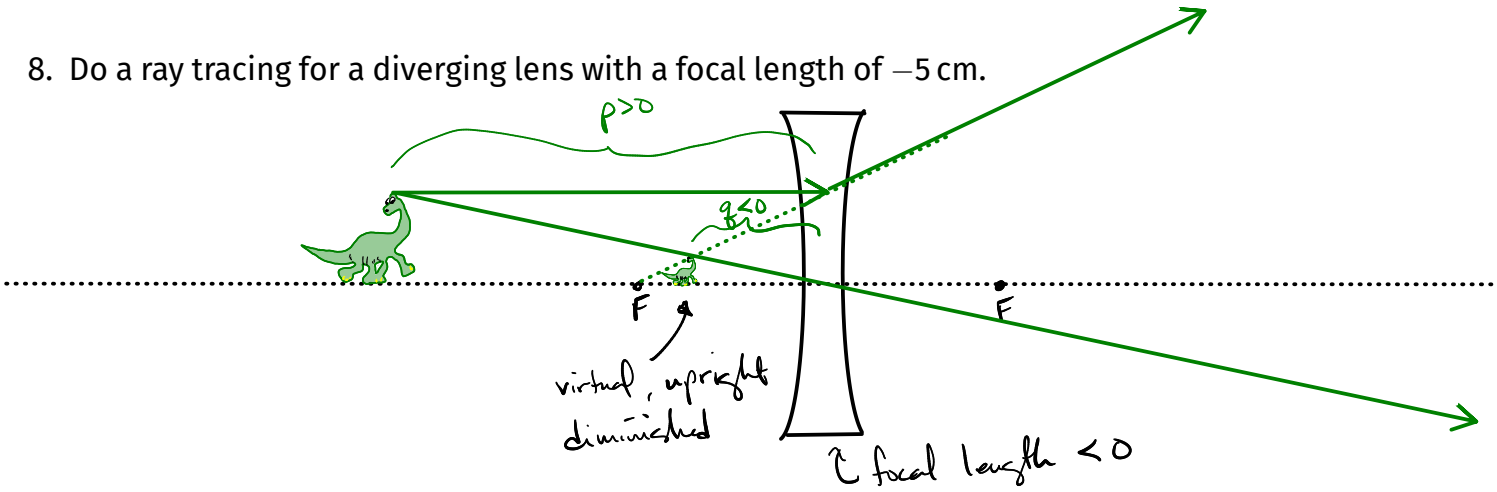
$$\frac{-q}{p} = m = \frac{h'}{h}$$

$$\frac{-q}{p} = -1$$

$$p = q$$



8. Do a ray tracing for a diverging lens with a focal length of -5 cm.



9. In order to magnify an object by 5 times and form a real image with a $+100$ mm focal length lens, where should the object be placed in front of the lens?

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{100}$$

$$\left(\frac{5}{5}\right)\frac{1}{p} + \frac{1}{5p} = \frac{1}{100}$$

$$-\frac{q}{p} = m = \frac{h'}{h}$$

$$-5$$

$$\frac{q}{p} = 5$$

$$q = 5p$$

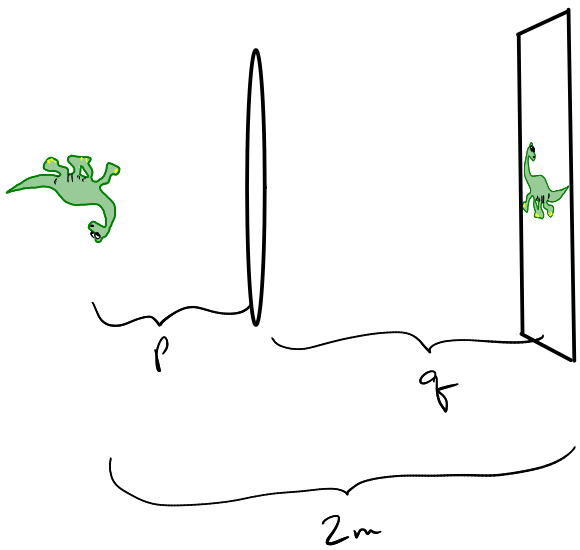
$$\frac{6}{5p} = \frac{1}{100}$$

$$\frac{600}{5} = p$$

$$\boxed{120\text{mm} = p}$$

$$\boxed{q = 600\text{mm}}$$

10. You make a work of art that consists of an object ~~X~~ 2m away from where you want a real image formed. Where should you place a 10 cm lens in order to make the image?

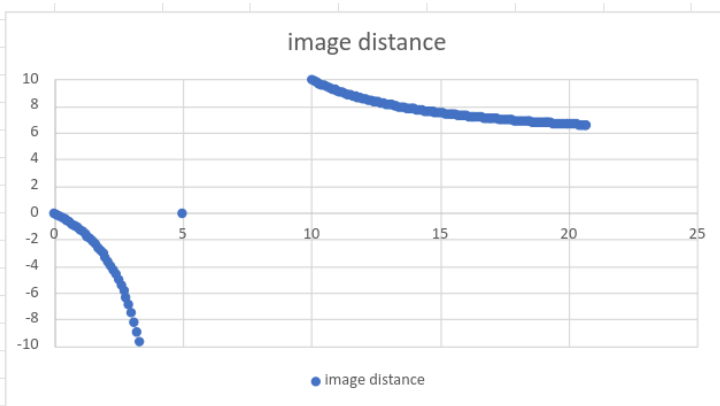


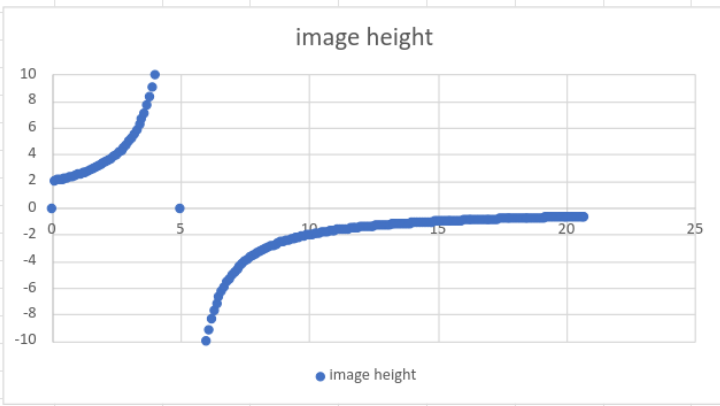
- $\frac{1}{p} + \frac{1}{q} = \frac{1}{0.10\text{m}}$
- $p + q = 2\text{m}$

$$q = 2 - p$$

$$\frac{(2-p)}{(2-p)}\frac{1}{p} + \frac{1}{(2-p)}\left(\frac{p}{p}\right) = 10$$

$$\frac{2-p}{p(2-p)} + \frac{p}{p(2-p)} = 10$$





$$\frac{2}{p(2-p)} = 10$$

$$\frac{2}{2p-p^2} = 10$$

$$\frac{2}{10} = \frac{10(2p-p^2)}{10}$$

$$0.2 = 2p - p^2$$

$$0 = -p^2 + 2p - 0.2$$

$$0 = ax^2 + bx + c$$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$p = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

choose for artistic reasons

$$\boxed{p = 0.1055 \rightarrow q = 1.89 \qquad m = -\frac{q}{p} = -17.9}$$

— or —

$$p = 1.89 \rightarrow p = 0.1055 \qquad m = -0.055$$

tiny image